

Jonathan Bucher

Software Developer · C++ / Graphics / Unity · Open Work Permit (Canada)
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EDUCATION

M.Sc. Informatics — Technical University of Munich

Apr 2023 – Jan 2026

Grade: 1.6/5.0 (German scale, 1.0 highest) · Thesis grade: 1.0 (highest possible)

- Thesis: Architected and optimized a voxel-based flood-fill algorithm using a chunk-based, data-oriented approach with Unity's C# Job System and Burst compiler
- Image Synthesis (3D Rendering), Geometry Processing, Parallel Programming

Exchange Semester — École Polytechnique de Montréal

Aug 2025 – Dec 2025

- Mixed Reality Software Development (Android/Unity, AR/VR), Intelligent Agents in Video Games (Unreal Engine)

Exchange Semester — Norwegian University of Science and Technology

Jan 2024 – Jun 2024

- Graphics and Visualization, Advanced Software Engineering, Advanced Database Management Systems

B.Sc. Informatics: Games Engineering — Technical University of Munich

Oct 2019 – Mar 2023

Grade: 1.9/5.0 (German scale, 1.0 highest)

- Thesis: Designed formal spatial representations for machine learning based AI players in a 4X strategy game using Unity's ML Agents
- Real-time Computer Graphics, C++ Programming, Operating Systems and Hardware-oriented Programming

KEY PROJECTS

Voxel-Based Spatial Partitioning System ([more info](#))

M.Sc. Thesis, TUM · 2025

- Architected a chunk-based parallel voxel flood-fill algorithm using Unity C# Jobs and Burst compiler
- Rewrote codebase to be fully Burst-compatible; optimized data layout for cache efficiency and memory footprint
- Achieved 71× peak speedup; reduced query complexity from linear to amortized O(1); highest possible grade

Autonomous Drone Navigation ([more info](#))

Mobile Robotics Practicals, TUM · 2025

- Built a full autonomous navigation pipeline in C++: camera projection, visual-inertial localization, sparse landmark matching, and feedback-controlled pathfinding under real-time frame-time constraints
- Completed the task in 25 seconds; fastest of 8 teams and well under the 40-second threshold for top grade

Le P'tit Alchimiste – VR Sandbox ([more info](#))

Meta Quest 3, Unity · Polytechnique Montreal · 2025

- Implemented a URP water shader with movement-based surface displacement for immersive VR interaction

C++/OpenGL Water Shader ([more info](#))

C++ · NTNU · 2024

- Implemented a stylized water shader from scratch in C++/OpenGL with incrementally toggled rendering stages

WORK EXPERIENCE

Software Engineering Intern — Itestra - Java, PostgreSQL, Elasticsearch, Spring Boot

Oct 2023 - Dec 2023

- Migrated a large-scale legacy retail system to a modern Spring Boot architecture within a small agile team
- Built reusable pipelines to migrate and synchronize millions of records across PostgreSQL and Elasticsearch, including data cleansing and transformation for complex legacy datasets

Full Stack Developer Intern — Medigital - React, Django, TypeScript

May 2023 – Oct 2023

- Developed features in front and backend of a web-based patient bed management system for hospitals

CORE TECHNICAL SKILLS

Languages: C++20, C#, HLSL/GLSL, Python, C, Java, SQL

Frameworks & Tools: Unity, Unreal Engine, Git, Visual Studio, Docker, Linux, Blender

Systems & Graphics: OpenGL, real-time rendering, HLSL shaders, parallel programming, SIMD

Software Engineering: Agile/Scrum, distributed systems, testing (JUnit, pytest)

Languages: German (native), English (fluent), French (B1), Norwegian (A1)